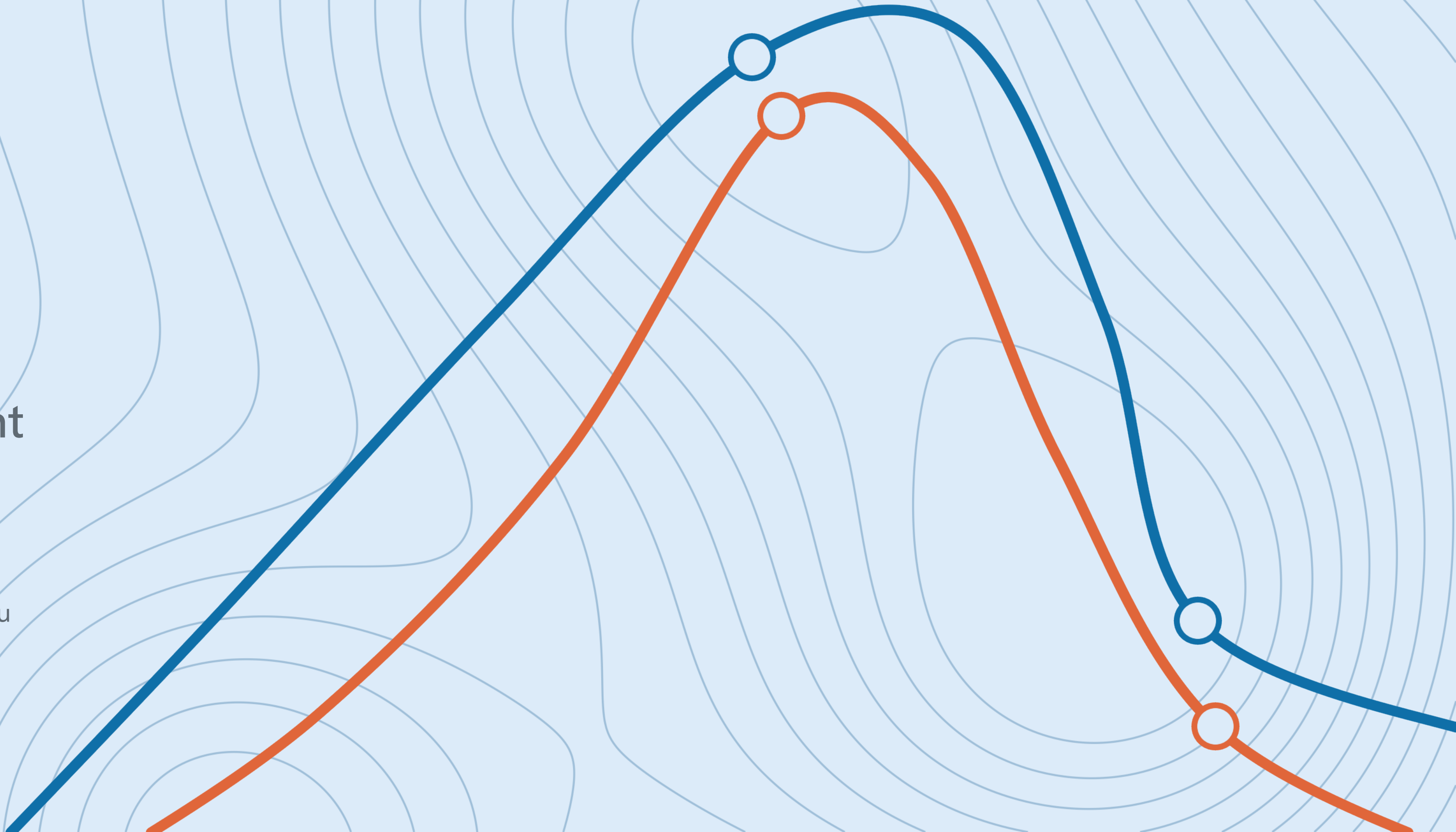


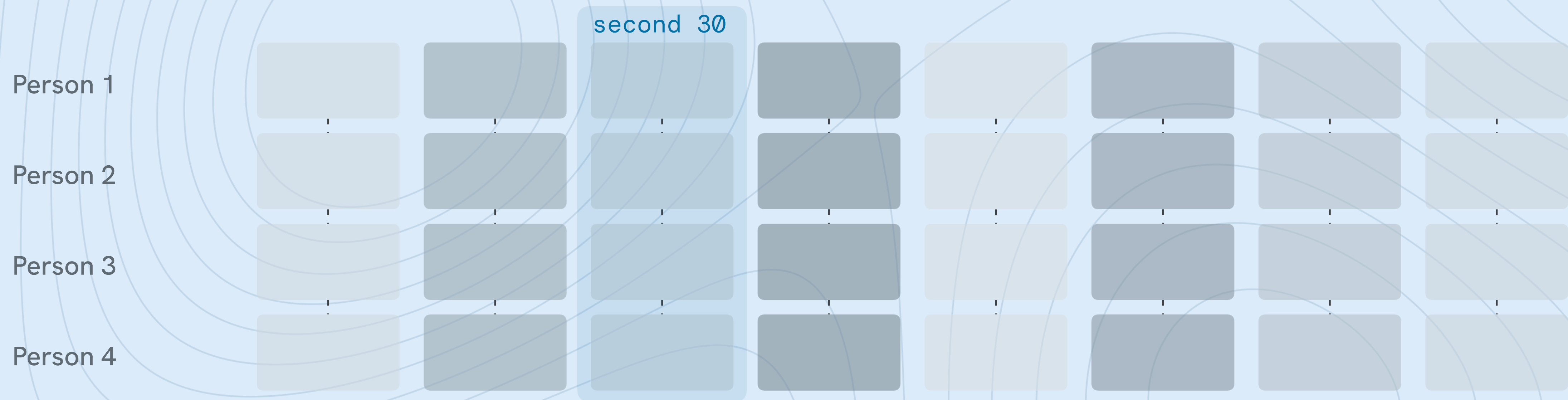
TrajOT Align

Knowing when a brain alignment
is real

Anand Lo · Het Jivani · Nafisah Nubah
Rafat Hossain · Zawad Atif · Sophie Eruokwu



Comparing brain scans usually needs a shared timeline



Movie: second 30 is the same scene for everyone

A Reduced-Dimension fMRI Shared Response Model

Po-Hsuan (Cameron) Chen, Janice Chen, Yaara Yeshurun, Uri Hasson, James Haxby, Peter J. Ramadge

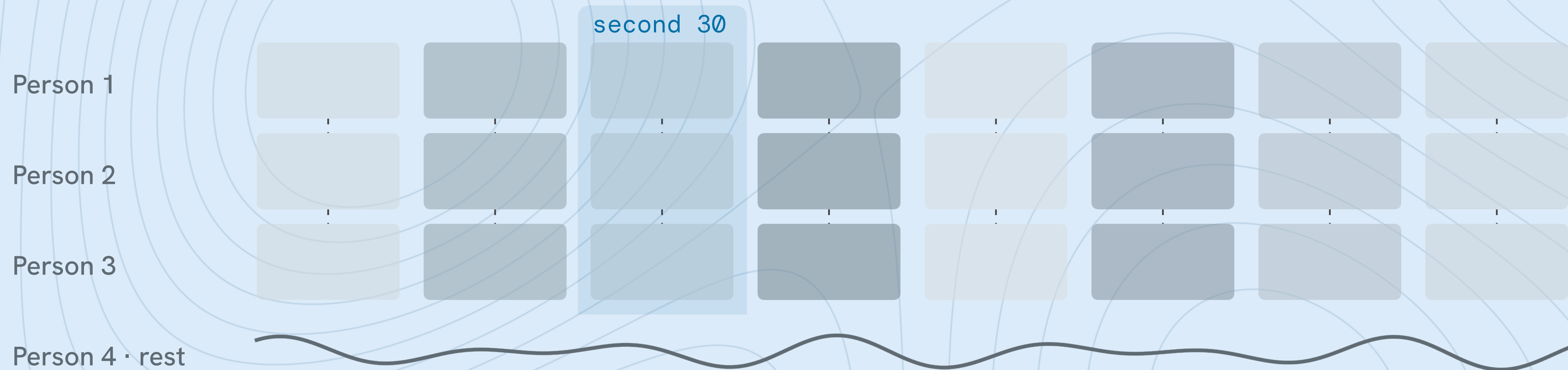
Advances in Neural Information Processing Systems 28 (NIPS 2015)

> [Neuron](#). 2011 Oct 20;72(2):404-16. doi: 10.1016/j.neuron.2011.08.026.

A common, high-dimensional model of the representational space in human ventral temporal cortex

James V Haxby¹, J Swaroop Guntupalli, Andrew C Connolly, Yaroslav O Halchenko, Bryan R Conroy, M Ida Gobbini, Michael Hanke, Peter J Ramadge

Comparing brain scans usually needs a shared timeline



Rest: no shared moment to line up

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Every method returns a map

An alignment that captured nothing still returns a map that **looks plausible**.

no uncertainty

no failure signal

group atlas $\neq$ individual

unvalidated per subject

"the results of unsupervised alignment at the individual level were statistically unreliable" – Takeda et al. 2025 [3]

What we align

Each dot = one patch of cortex, placed by how it connects to every other patch.

The map = which patch in one person matches which in another.

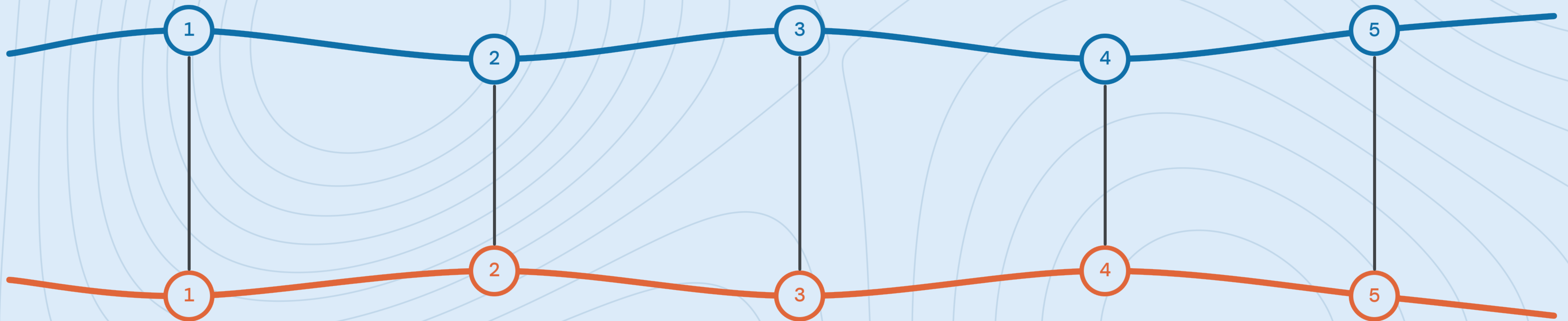
Everyone is matched to one shared template, so pairwise maps come from one model, not one fit per pair.

5,124 surface vertices → 100 Schaefer regions. Shown here in 2-D.

[illustration – real embedding from the method team]

Existing methods return one map

Person A

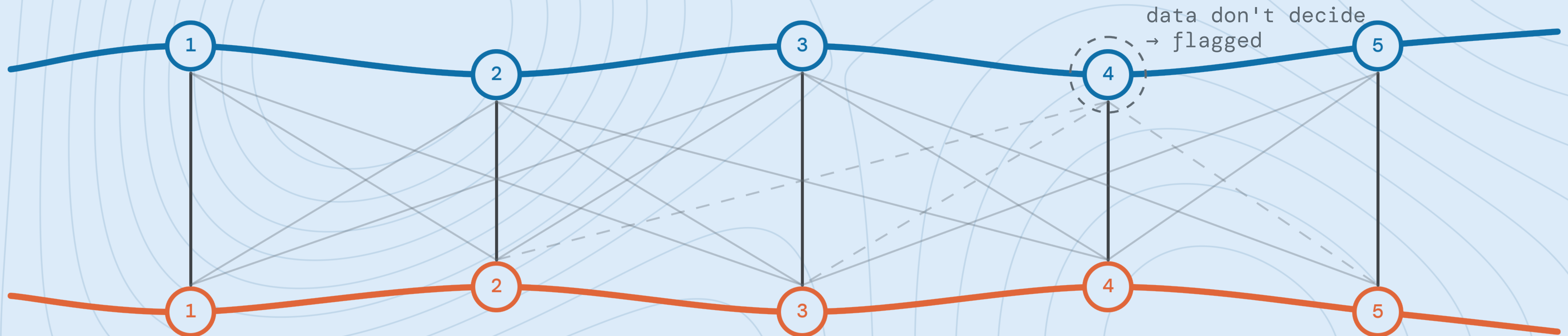


Person B

BrainSync, FUGW, connectivity-SRM: one map, no error bar

TrajOT returns a distribution

Person A



Person B

TrajOT: a distribution over maps — its width says whether the data decide

What existing tools cover

Method	Returns a map for every pair?	Reports uncertainty?	Posterior over maps?
BrainSync	Yes	No	No
FUGW	Yes	No	No
connectivity-SRM	Yes	No	No
GW toolbox (GWTune)	Yes	No	No
TrajOT	Yes	Yes, per subject	Yes

Every method returns a map for every pair.
None says which ones to trust.

ARTICLE

BrainSync: An Orthogonal Transformation for Synchronization of fMRI Data Across Subjects

Authors: Anand A. Joshi, Mingi Chong, Richard M. Leahy | [Authors Info & Claims](#)

Medical Image Computing and Computer Assisted Intervention – MICCAI 2017: 20th International Conference, Quebec City, QC, Canada, September 11-13, 2017, Proceedings, Part I
Pages 486 - 494 • https://doi.org/10.1007/978-3-319-66182-7_56

Published: 10 September 2017 [Publication History](#)

> PLoS Comput Biol. 2018 Apr 17;14(4):e1006120. doi: 10.1371/journal.pcbi.1006120. eCollection 2018 Apr.

A computational model of shared fine-scale structure in the human connectome

J Swaroop Guntupalli ^{1 2 3}, Ma Feilong ^{1 2}, James V Haxby ^{1 2}

Affiliations + expand

PMID: 29664910 PMCID: [PMC5922579](#) DOI: [10.1371/journal.pcbi.1006120](#)

Quantitative Biology > Neurons and Cognition

[Submitted on 19 Jun 2022 (v1), last revised 22 Aug 2023 (this version, v3)]

Aligning individual brains with Fused Unbalanced Gromov-Wasserstein

Alexis Thual, Huy Tran, Tatiana Zemskova, Nicolas Courty, Rémi Flamary, Stanislas Dehaene, Bertrand Thirion

The framework

steps 1-3

1 · Connectomes

Each run becomes a 100×100 Fisher-z connectome over Schaefer parcels. Intra-subject geometry is the signal.

2 · Features

Diffusion-map coordinates plus curvature and sulcal depth. Surface geodesics give the anatomical anchor M^0 .

3 · Calibrate β

$\beta = \sigma C^{-2}$ from how much a connectome moves between two same-day scans. 28.44 here — measured, not tuned.

Rest has no shared timeline, so the model learns from each brain's own internal geometry.

$K = 100$ template nodes · Schaefer-100 · rank 32 · 30 Sinkhorn steps · ϵ annealed 0.1 → 0.01

The framework

steps 4-6

4 · Posterior

Encoder scores parcels against template nodes; perturb and Sinkhorn-project, so every draw is a valid plan.

5 · Train

β -weighted Gromov-Wasserstein, a feature term that cuts GW's isometry symmetry, an entropic prior on M^0 .

6 · Apply + pool

$\tilde{C} = (1-\lambda)C + \lambda Q^T C Q$, λ from the posterior's row-entropy: uncertain subjects move less. Group REML down-weights them again.

The posterior drives the map.

K = 100 template nodes · Schaefer-100 · rank 32 · 30 Sinkhorn steps · ε annealed 0.1 $\rightarrow$ 0.01

How we tested it

Resting scans

OpenNeuro ds000243
N = 83 two-run · TR =
2.5 s



Geometry

Schaefer-100 regions
100 × 100 connectome



Align

TrajOT and each
baseline
same map Q on both
runs



The check

scan-rescan
reliability:
does a person's own
two scans
agree better after
alignment?

Primary metric: scan-rescan reliability · 10,000 paired bootstrap, McNemar exact.

COMPARED AGAINST · same preprocessing, folds and evaluation code

A person's two scans agree better

Scan-rescan reliability, TrajOT minus each baseline. 95% CIs, 10,000 paired bootstrap.

vs no alignment

+0.0519

vs BrainSync

+0.0513

vs FUGW

+0.0759

**All three intervals
exclude zero.**

Reliability after
alignment
TrajOT 0.698
no alignment 0.646
BrainSync 0.646
FUGW 0.622

N = 83 two-run subjects · ds000243 · Schaefer-100 · $\beta = 28.44$

The protocol is map-invariant — a Haar-random map scores 83/83 too.

What it costs, and what we don't claim

Identification

Under one map applied to both runs, identification is map-invariant: a Haar-random map also scores 83/83. It cannot rank aligners here.

Cross-subject gain

Ours is -0.035 . Pulling a subject toward the template costs cross-subject correlation on this protocol.

conn-SRM

$+0.395$ gain, but identification collapses to chance: the fingerprint is erased, not aligned.

Only the hierarchical model reports uncertainty: $\tau\phi = 0.0345$, nine subjects flagged at the 90th percentile, and group REML with $n_{\text{eff}} < S$. Baselines leave every one of those columns empty.

What we tried and what we learned

Tried	What happened	What we'd do differently
Identification as the headline metric	Under a same map it is map-invariant — a Haar-random map scores 83/83	Score reliability, and pre-register a protocol that can rank maps
Procrustes toward the population mean	Point path: +0.018 gain, ident 0.857 — superseded by the posterior path	Let the posterior set how far each subject moves
BrainSync on connectomes	A structural no-op — feature corr 0.998	Compare on time series, not connectomes
Counting non-identifiable pairs	Null degenerate at $B = 200$; counts saturate 466–500	Define it from posterior width, not the gain null
Applying the posterior to the map	Wired: λ comes from posterior row-entropy, λ mean 0.50	Calibrate coverage next — it still fails on real rest

$N = 83 \cdot 10,000$ paired bootstrap

Using it

```
git clone https://github.com/HetJivani04/\
  SURGE-Neurohack-DMLS-Fall-2026.git
pip install -e trajot/
cp configs/paths.example.yaml configs/paths.yaml
python scripts/run_experiment.py \
  --config configs/experiments/10_ours_full.yaml
```

```
runs/<run_id>/
manifest.json  config + hash, git
               commit, data hash, seed, versions
metrics.json  ident, gain, p-values,
                $\tau\phi$ 
log.txt       stdout and loss traces
artifacts/    couplings,  $\beta$ , figures
```

Re-runnable from manifest.json alone.

CPU-only, float64 on a 16 GB laptop

[github.com/HetJivani04/\
 SURGE-Neurohack-DMLS-
 Fall-2026](https://github.com/HetJivani04/SURGE-Neurohack-DMLS-Fall-2026)



Next steps





Next directions

Patient groups who can't do scanner tasks

e.g. stroke, aphasia, young children

**Comparing patients with each other, and
with themselves across visits**

Combining resting data across hospitals

**Carrying alignment uncertainty into
group statistics**

so unsure maps count less

References

- [1] Haxby et al. 2011 · Neuron
- [2] Chen et al. 2015 · NeurIPS (SRM)
- [3] Takeda et al. 2025 · iScience
- [4] Joshi et al. 2018 · NeuroImage (BrainSync)
- [5] Thual et al. 2022 · NeurIPS (FUGW)
- [6] Nastase et al. 2020 · NeuroImage (connectivity-SRM)
- [7] Mazelet, Flamary & Thirion 2025 · NeurIPS (ULOT)
- [8] Keller et al. 2008 · Statistica Sinica
- [9] Finn et al. 2015 · Nat Neurosci (identification)
- [10] Takeda et al. 2025 · J Neurosci Methods (GWTune)
- [11] Schaefer et al. 2018 · Cereb Cortex (atlas)
- [12] ds000243 · Power et al. 2017 · PLOS ONE

Full reference list in our repo.